Introduction

Partial vs general equilibrium–Example

- Partial equilibrium analysis valid when applied to small-size markets; Vives (1987) shows a market is ``small” if
  - Wealth effects are small: each consumer spends small fraction of income on the good.
  - Dispersed substitution effects: prices of other goods are essentially unaffected by changes in given market.

Somewhat arbitrary literature classification

	Price-taking	**Strategic behavior**
Partial eq.	Consumer/Producer theory	GT/IO/oligopoly
General eq.	Arrow-Debreu economies	

Adam Smith (Smith 1776)

… every individual necessarily labours to render the annual revenue of the society as great as he can. He generally, indeed, neither intends to promote the public interest, nor knows how much he is promoting it. By preferring the support of domestic to that of foreign industry, he intends only his own security; ***and by directing that industry in such a manner as its produce may be of the greatest value, he intends only his own gain, and he is in this, as in many other cases, led by an invisible hand to promote an end which was no part of his intention.*** *Nor is it always the worse for the society that it was no part of it.* ***By pursuing his own interest he frequently promotes that of the society more effectually than when he really intends to promote it.*** *I have never known much good done by those who affected to trade for the public good.*

Adam Smith

Laws and government may be considered in this and indeed in every case as a combination of the rich to oppress the poor, and preserve to themselves the inequality of the goods which would otherwise be soon destroyed by the attacks of the poor, who if not hindered by the government would soon reduce the others to an equality with themselves by open violence.

Tuesday, February 2nd, 1763.

Lectures in Jurisprudence (1762-63)

Walrasian equilibrium, Pareto optimality

A Walrasian equilibrium (WE) (or competitive equilibrium) is a collection of prices, and an assignment of goods to consumers and of production plans to firms such that

- for every consumer, the allocated bundle maximizes utility subject to the budget constraint given prices,
- for every firm, the allocated production plan maximizes given prices,
- markets clear, i.e. demand = supply.

An allocation for consumers and a production plan for firms is **Pareto optimal (PO)** if it is not possible to rearrange production and consumption to make everyone better off. (Pareto 1896, 1897).

Main questions

1. Welfare properties of WE

(In favor of market economies: outcomes are WE and WE are PO.)

- **First Welfare Theorem**. Every WE is Pareto optimal (Arrow-Debreu).
- True in great generality.
- Main assumption: price-taking behavior in definition of WE.
- Fails under externalities, incomplete markets, overlapping generations.

and

(WE do not favor certain PO allocations)

- **Second Welfare Theorem.** Every Pareto optimal allocation is a WE provided income is appropriately redistributed. (Debreu 1959)
- Even if PO cannot be computed, PO may be achieved by redistributing income and using FWT.
- Requires convexity of preferences and technology. Approximate results without convexity and large economies.
- Needs lump sum taxes for income transfers.

2. Existence

- Consistency.
- Counting equations and unknowns. (Walras 1874, 1874b),
- Scientific approach. (Wald 1934, 1935).
- (Arrow and Debreu 1954), (Debreu 1959), (McKenzie 1954, 1954b, 1980).

  Mathematical techniques:

  - von Neumann's Lemma.
  - Fixed Point Theorem (Kakutani 1941).
  - existence of Nash equilibrium (Nash 1950).

3. Uniqueness and policy.

- Comparative statics.
- Uniqueness: Is there a unique WE?
- Determinacy.
- Hildenbrand, Grandmont and Quah.
- For most economies, there are finitely many equilibria and the equilibria move smoothly with the parameters of the economy (Debreu 1970).

4. Price-taking assumption: Core-WE

- Edgeworth argued outcomes of market economies are allocations that exhaust all gains from trade among agents and groups of agents (Edgeworth 1881)).

If those outcomes are ``nearly” WE, then the price-taking assumption in the definition of WE is justified.

- Core convergence and core equivalence (Shubik 1959), (Debreu and Scarf 1963), (Aumann 1964), (Anderson 1978), (Manelli 1991).

Conclusion

- GE model as benchmark.
- Achievements and methodology

1. Successful model of exchange: given endowments (or production and income) explains distribution.
2. Less successful model with production: firms, goods, technology are exogenous; no markets for shares. (Perhaps should be strategic?)
3. How to analyze effects of policy changes such as taxes.

- Competitive applications: Finance, freshwater macro
- Deviations
  - Distortions from monopoly or oligopoly: Industrial organization
  - Distortions from taxes: Public finance
  - Effects of welfare payments or taxes on labor supply: Labor economics
  - Effect of Externalities

Preliminaries

Notation, definitions

- Let $r \in R$, $N \in \mathbb{N}$, $x = (x_1, \dots, x_N) \in \mathbb{R}^N$ (coordinates in x are indexed by $n = 1, \dots, N$).

$$
\begin{aligned}
x &= (x_1, \dots, x_N). \\
x_{-n} &= (x_1, \dots, x_{n-1}, x_{n+1}, \dots, x_N). \\
(x_{-n}, r) &= (x_1, \dots, x_{n-1}, r, x_{n+1}, \dots, x_N).
\end{aligned}
$$

- $\forall x, y, z \in \mathbb{R}^N$,

$$
\begin{array}{lll}
x \geq y & \text{if} & (\forall n)\ x_n \geq y_n. \\
x > y & \text{if} & x \geq y \ \wedge\ (\exists n)\ x_n > y_n. \\
x \gg y & \text{if} & (\forall n)\ x_n > y_n.
\end{array}
$$

- Let A_1, A_2 be subsets of a vector space X. Then,
- $A_1 + A_2 = \{x_1 + x_2 : x_1 \in A_1, x_2 \in A_2\}$.
- $\operatorname{co} A_1$ is the convex hull of A_1
- Lemma. If A_1, A_2 are convex, then $A_1 + A_2$ is convex.
- For $x \in \mathbb{R}^L$, $\|x\|_\infty = \max\{|x_\ell| : 1 \leq \ell \leq L\}$.

Binary relations

Binary relations are used to formalize preferences.

Definition. A *binary relation* R on the set X is a set $R \subseteq X \times X$. One writes $x \, R \, y$ to indicate that $(x, y) \in R$.

- A binary relation R on a set X is
  - complete if $\forall x, y \in X$ with $x \neq y$, $x \, R \, y$ or $y \, R \, x$.
  - reflexive if $\forall x \in X$, $x \, R \, x$.
  - irreflexive if $\forall x \in X$, $\neg(x \, R \, x)$.
  - transitive if $\forall x, y, z \in X$, $x \, R \, y \wedge y \, R \, z \implies x \, R \, z$.
  - symmetric if $\forall x, y \in X$, $x \, R \, y \implies y \, R \, x$.
  - asymmetric if $\forall x, y \in X$, if $x \, R \, y \implies \neg(y \, R \, x)$.
  - antisymmetric if $\forall x, y \in X$, if $x \, R \, y \wedge y \, R \, x \implies x = y$.
  - an equivalence relation if it is reflexive, symmetric, and transitive.

Binary relations

- Let $\succeq$ be complete and transitive.
  - Indifference: $x \sim y$ if $x \succeq y$ and $y \succeq x$.
  - Strict preference: $x \succ y$ if $x \succeq y$ and $\neg(y \succeq x)$.
- The following properties require additional structure on X.
  - continuous if $\{(x, y) \mid x \succ y\}$ is relatively open in $X \times X$.
  - convex if $\{x \in X : x \succ y\}$ is convex $\forall y \in X$.
  - strongly (or strictly) convex if it is convex and $x, y \in X$, $(x \neq y \ \wedge \ x \sim y) \implies \frac{x+y}{2} \succ x$.

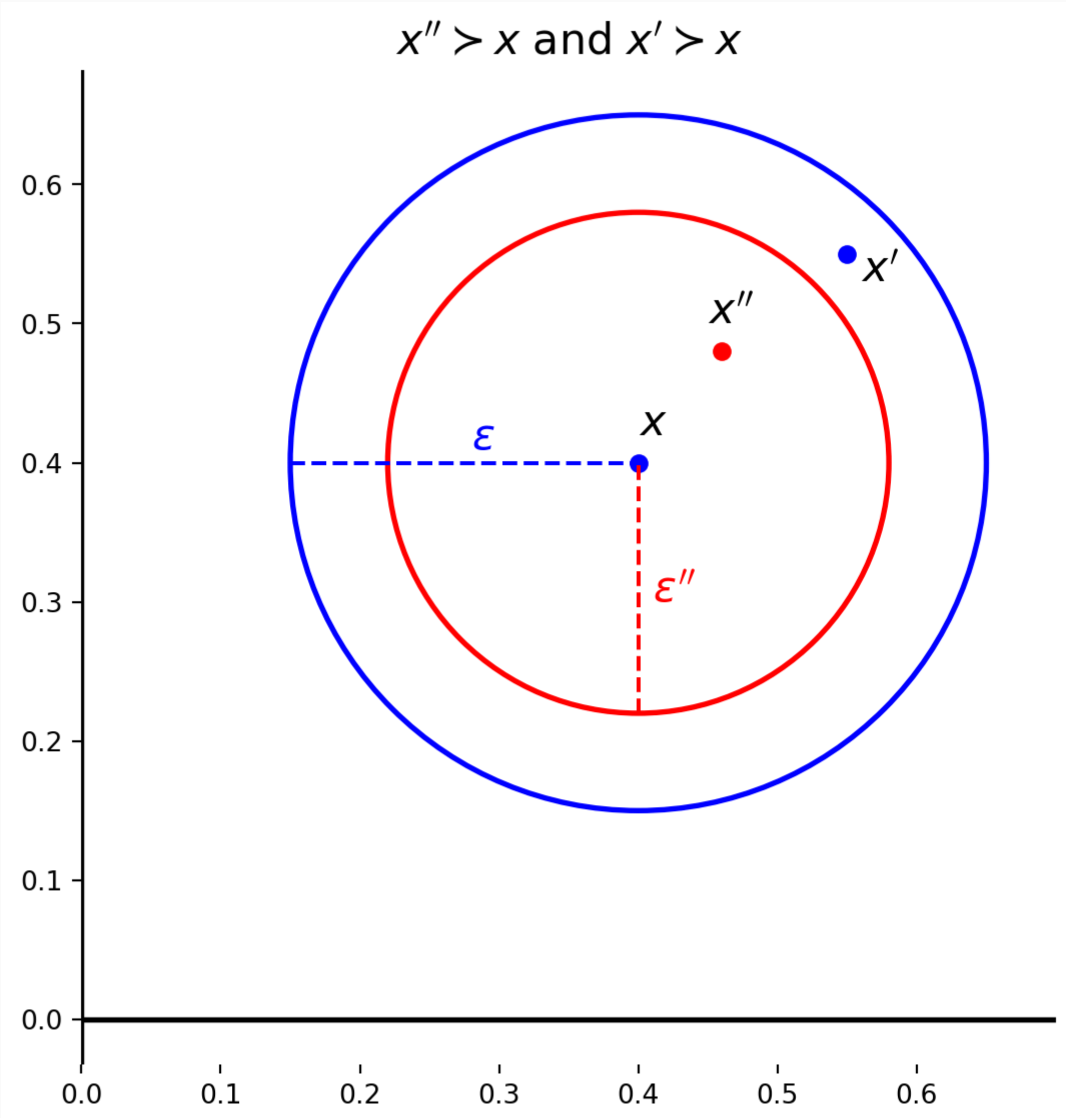
Binary relations

- the binary relation $\succ$ is
  - strongly monotone if $x > y \implies x \succ y$
  - (weakly) monotone if $x \gg y \implies x \succ y$
  - locally nonsatiatied (LNS) if $(\forall x \in X \wedge \forall \epsilon > 0)$,

$$\exists x' \in X : (\|x' - x\| < \epsilon \wedge x' \succ x).$$

LNS is a weakening of monotonicity. Important in economies with production because many inputs are not consumed.

LNS



$x'' \succ x$ and $x' \succ x$
0.0
0.1
0.2
0.3
0.4
0.5
0.6
x'
x''
x
ε
ε''

Properties of preferences

- Let $\succeq$ be a complete, transitive, continuous and LNS relation on a closed subset X of $\mathbb{R}^L$. Then X must be unbounded.

Technology, production sets

- A production plan is a vector in a production set, $y = (y_1, y_2, \dots, y_L) \in Y \subset \mathbb{R}^L$.
- Convention: inputs are negative and outputs are positive.
- Reason: $p \cdot y$ can be interpreted as profit.
- Example. $Q = F(L)$, $Q = 2L$. Let $Q = y_2$ and $-L = y_1$.

Model

The Arrow-Debreu economy

Definition 1 Let $L, J, I \in \mathbb{N}$ be the number of goods, firms, and agents; indexed by ℓ, j and i respectively. An Arrow-Debreu economy is a collection $\mathcal{E} = \left(L, \{Y_j\}_{j=1}^{J}, \{X_i, \succeq_i\}_{i=1}^{I}, \left\{\omega_i, \{\theta_{ij}\}_{j=1}^{J}\right\}_{i=1}^{I}\right)$ where

- $Y_j \subseteq \mathbb{R}^L$ is the technology available to firm j,
- $X_i \subseteq \mathbb{R}^L$, is $i's$ consumption set.
- $\succeq_i \subseteq X_i \times X_i$ is i's preference.
- $\omega_i \in \mathbb{R}^L$ is i's endowment;
- $\theta_{ij} \in [0, 1]$ is i's share of firm j's profit and $\sum_{i=1}^{I} \theta_{ij} = 1$.

Pure exchange economies

Definition 2 An Arrow-Debreu economy where there is no production, i.e. $\mathcal{E} = \left(L, \{X_i, \succeq_i\}_{i=1}^{I}, \{\omega_i,\}_{i=1}^{I}\right)$ is called a pure exchange economy or simply an exchange economy.

- Endowments and shares determine the economy's ownership structure in an economy with production. In a pure exchange economy the ownership structure is given by the distribution of endowments.

Commonly used assumptions about firms

- $0 \in Y_j$
- Y_j is closed
- $\mathbb{R}^L_+ \cap Y_j = \{0\}$
- $Y_j + Y_j = Y_j$ (Additivity)
- $Y_j - \mathbb{R}^L_+ \subseteq Y_j$ (Free disposal) or slightly weaker $-\mathbb{R}^L_+ \subseteq Y_j$.
- Y_j is convex

Commonly used assumptions about agents

- Consumption sets X_i are subsets of $\mathbb{R}_+^L$.
- Preferences $\succeq_i$ are a binary relation on X_i that are
  - complete,
  - reflexive,
  - transitive,
  - continuous,
  - convex.

Definitions

Fix an Arrow-Debreu economy $\mathcal{E}$.

Definition 3 The aggregate endowment in $\mathcal{E}$ is $\bar{\omega}$ where $\bar{\omega} = (\bar{\omega}_1, \dots, \bar{\omega}_L)$ $= \sum_{i=1}^{I} \omega_i$.

Definition 4 A pair $(x, y) = (x_1, \dots, x_I, y_1, \dots, y_J)$ in $(\mathbb{R}^L)^I \times (\mathbb{R}^L)^J$ is an allocation if

- $(\forall i) \quad x_i \in X_i$;
- $(\forall j) \quad y_j \in Y_j$; and
- $\sum_{i=1}^{I} x_i = \sum_{i=1}^{I} \omega_i + \sum_{j=1}^{J} y_j$ (feasibility).

Pareto optimality

Definition 5 (Pareto optimal allocation) An allocation (x, y) is

- (strongly) Pareto optimal if there is no allocation (x', y') such that $\forall i, x'_i \succeq_i x_i$ and $(\exists i)\ x'_i \succ_i x_i$;
- (weakly) Pareto optimal if there is no allocation (x', y') such that $\forall i, x'_i \succ_i x_i$.

If there is such an allocation (x', y'), we say that (x', y') (weakly/strongly) Pareto dominates (x, y).

Budget and Demand sets

Definition 6 (Budget set) Given $p \in \mathbb{R}^L$, $y \in \prod_{j=1}^J Y_j$, and a transfer $t_i \in \mathbb{R}$, agent i's budget set is

$$B_i(p, y, \omega_i, t_i) = \{x_i \in X_i : p \cdot x_i \leq p \cdot \omega_i + t_i + \sum_{j=1}^{J} \theta_{ij}(p \cdot y_j)\}.$$

Definition 7 (Demand set) Given $p \in \mathbb{R}^L$, $y \in \prod_{j=1}^J Y_j$, and a transfer $t_i \in \mathbb{R}$, agent i's demand set is

$$D_i(p, y, \omega_i, t_i) = \{x_i' \in B_i(p, y, \omega_i, t_i) : \forall x_i \in B_i(p, y, \omega_i, t_i), \;\; x_i' \succeq_i x_i\}.$$

Walrasian Equilibrium

Definition 8 (Walrasian equilibrium) $(x^*, y^*, p^*, t^*) \in \mathbb{R}^{IL} \times \mathbb{R}^{JL} \times \mathbb{R}^L \times \mathbb{R}^I$ is a WE with transfers of the Arrow-Debreu economy $\mathcal{E}$ if

1. $p^* \in \mathbb{R}^L$, $t^* = (t_1^*, \dots, t_I^*)$ and $\sum_{i=1}^I t_i^* = 0$,
2. $(\forall j)\ \left[y_j^* \in Y_j \wedge (\forall y_j \in Y_j)\ p^* \cdot y_j^* \geq p^* \cdot y_j\right]$ (price-taking profit maximization)
3. $(\forall i)\ x_i^* \in D_i(p^*, y^*, \omega_i, t_i^*)$ (price-taking preference maximization)
4. $\sum_{i=1}^I x_i^* = \sum_{i=1}^I \omega_i + \sum_{j=1}^J y_j^*$ (market clearing), i.e. (x^*, y^*) is an allocation.

If $(\forall i)\ \ t_i^* = 0$, then (x^*, y^*, p^*) is called a WE.

Remarks

- Assumption that firms maximize profit? Assuming complete markets and complete information: If share holders decide, they agree on profit maximization. (Mas-Colell, Whinston, and Green 1995, chap. 5).
- Policy argument in favor of market economies has problems. Is a WE the outcome of a market economy?

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